Figure 1: **Capability Overview of Sync-R1.** Our framework jointly optimizes personalized understanding and generation within a unified reasoning trajectory via Sync-GRPO. By establishing an explicit synergistic loop, Sync-R1 leverages the reciprocal enhancement between comprehension and creation to achieve precise integration of personalized concepts. Beyond standard personalization, Sync-R1 excels in challenging scenarios, including dense-text and reasoning-based generation tasks with missing concept information.

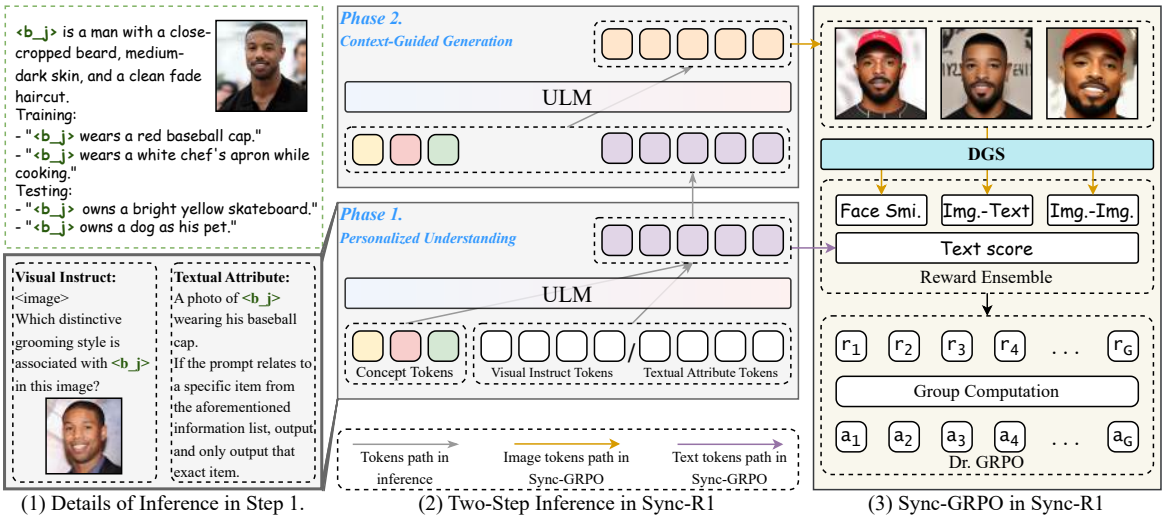

Figure 2: **Overview of the Sync-R1 Framework.** We introduce a novel integration of personalized understanding and generation reasoning. Through our proposed **Sync-GRPO**, these two processes are coupled to mutually reinforce each other, jointly contributing to parameter updates. This mechanism significantly enhances the model's reasoning capabilities regarding personalized concepts and achieves high-fidelity information injection.

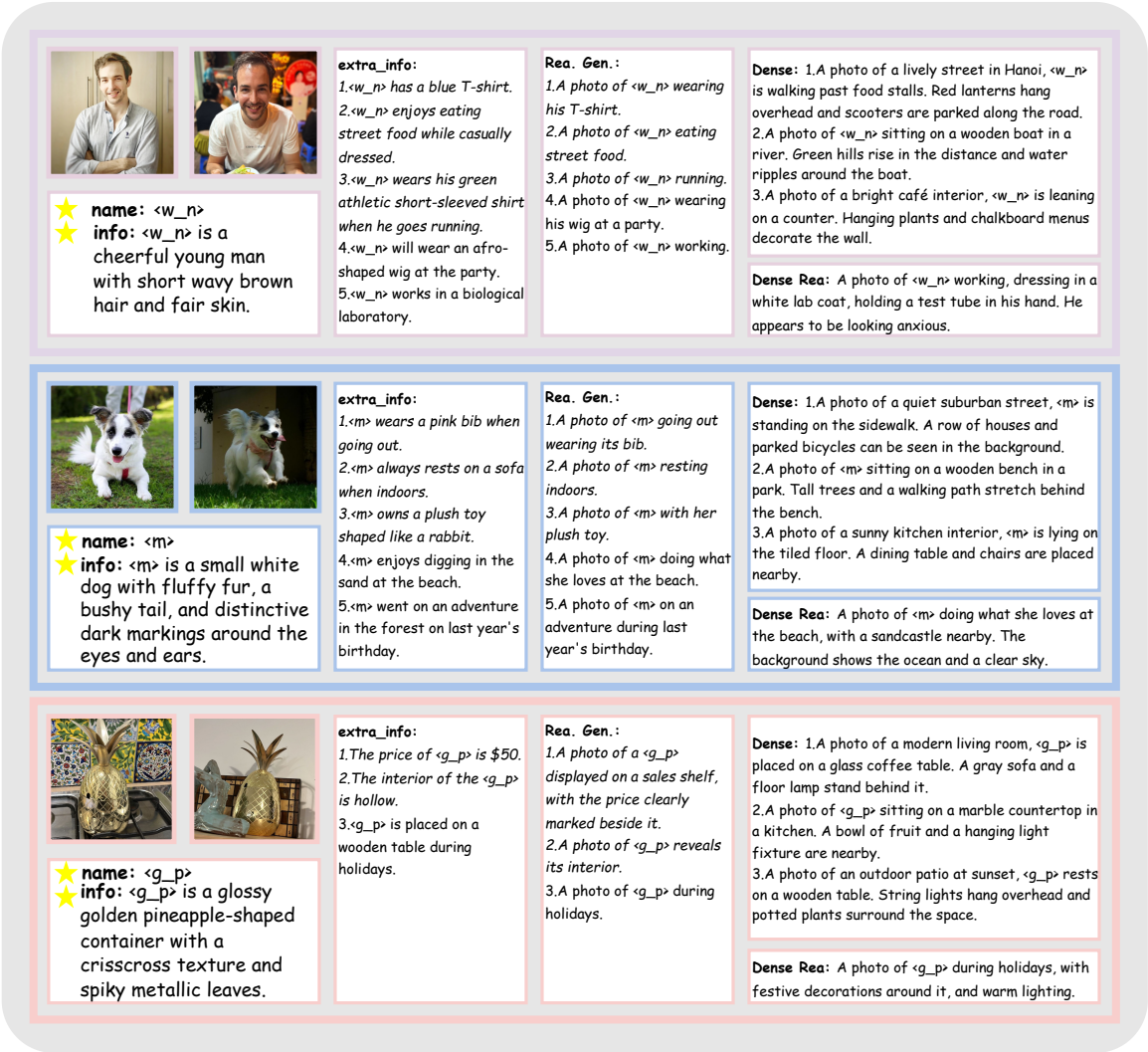

Figure 3: **Snapshot of UnifyBench++.** We present illustrative examples of concept entries within our constructed dataset. The *italicized segments* within the `extra_info` and `Rea. Gen.` columns denote the specific data subsets utilized for training Sync-R1, while the non-italicized text is reserved for evaluation.

Table 1: **Sensitivity of the reward-weighting strategy.** We perturb the overall weight for the understanding task ($w_{\text{Und.}}$) by $\pm 15\%$ around the default setting, while the remaining weight ($w_{\text{Gen.}}$) is equally distributed among all the generation reward functions. The relative changes across all averaged task scores are strictly less than 1%, demonstrating that Sync-R1 is highly robust to moderate coefficient perturbations and does not rely on fragile numerical tuning.

Weight Setting ($w_{\text{Und.}} : w_{\text{Gen.}}$)	Und.					Gen.			
	Rec.	Rea.*	Dense Rea.*	VQA	QA	Pure Gen.	Dense Gen.*	Rea. Gen.*	Dense Rea. Gen.*
	Weight	BLEU	GPT	Avg. Score	Avg. Score	Avg. Score	Avg. Score	Avg. Score	Avg. Score
+15% Und. (0.46:0.54)	0.853 (-0.698%)	0.252 (+0.800%)	0.499 (-0.795%)	0.599 (0.000%)	0.630 (+0.318%)	0.535 (-0.372%)	0.489 (-0.407%)	0.565 (+0.355%)	0.557 (+0.360%)
-15% Und. (0.34:0.66)	0.864 (+0.582%)	0.249 (-0.400%)	0.499 (-0.795%)	0.600 (+0.167%)	0.632 (+0.637%)	0.542 (+0.931%)	0.490 (-0.204%)	0.562 (-0.178%)	0.553 (-0.360%)
Sync-R1 (0.40:0.60)	0.859	0.250	0.503	0.599	0.628	0.537	0.491	0.563	0.555

Table 2: **Summary of the reward components in Sync-R1.** Each reward supervises a different aspect of the joint reasoning-generation trajectory. TIER is used for the understanding phase, whereas BER, DER, and FER evaluate generation fidelity. The weights are coarse category-level defaults rather than per-concept tuning.

Component	TIER	BER	DER	FER
Supervises	Understanding phase	Generation phase	Generation phase	Generation phase
Role	Understanding	Semantic Alignment	Identity Preservation	Facial Fidelity
Input	Intermediate reasoning output & candidate concept information	Text prompt & generated image	Generated image & reference image	Aligned face crops from generated and reference images
Target behavior	Encourage correct selection and grounding of relevant personalized concept information	Enforce prompt-image consistency and reduce semantic drift	Preserve concept-specific visual identity	Preserve subtle facial identity cues
Activated category	All categories	All categories	All categories	Human only
Weight (Human subjects)	0.4	0.2	0.2	0.2
Weight (Animals/Objects)	0.4	0.3	0.3	0.0

Table 3: **Comparison with larger-capacity contemporary baselines on Unifybench++.** We compare Sync-R1 (1.3B) with larger 7B-scale baselines, including Chameleon, Janus-Pro, and BAGEL. TP/IP denote text-prompt/image-prompt conditioning when supported by the baseline. Columns with * are newly extended from Unifybench. Best and second best results are highlighted; ties are marked with the same highlight.

Method	Model Size	Und.							Gen.							
		Rec.	Rea.*	Dense Rea.*	VQA		QA		Pure Gen.		Dense Gen.*		Rea. Gen.*		Dense Rea. Gen.*	
		Weight	BLEU	GPT	BLEU	GPT	BLEU	GPT	CLIP-T	CLIP-I	GPT	CLIP-I	CLIP-T	CLIP-I	GPT	CLIP-I
Chameleon+TP	7B	0.685	0.206	0.313	0.411	0.489	0.509	0.560	0.186	0.542	0.281	0.463	0.193	0.516	0.309	0.549
Chameleon+IP	7B	0.497	0.216	0.374	0.446	0.497	0.411	0.532	0.165	0.514	0.266	0.449	0.190	0.532	0.299	0.503
Janus-Pro+TP	7B	0.822	0.297	0.550	0.492	0.629	0.444	0.650	0.249	0.627	0.387	0.607	0.243	0.650	0.254	0.642
Bagel+TP	7B	0.856	0.278	0.500	0.391	0.641	0.366	0.600	0.284	0.643	0.317	0.628	0.263	0.658	0.349	0.638
Bagel+IP	7B	0.674	0.242	0.550	0.377	0.609	0.380	0.515	0.293	0.690	0.391	0.595	0.285	0.666	0.371	0.630
Sync-R1	1.3B	0.859	0.250	0.503	0.592	0.606	0.604	0.652	0.308	0.765	0.337	0.645	0.324	0.801	0.353	0.756

Table 4: **Wall-clock latency per rejected candidate in DGS.** The intermediate surrogate evaluation adds only 0.32s after the first 10 denoising steps, but allows DGS to skip 9.03s of subsequent decoding and final reward evaluation for low-potential trajectories. All latencies are averaged across 10 samples per concept.

Operation Phase	**Wall-clock Latency**	**Status in DGS (for rejected samples)**
1. Initial Decoding (Steps 0–9)	$\sim$ 2.12 s	Executed
2. Intermediate VQ & BLIP Scoring	$\sim$ **0.32 s**	**Executed (Overhead)**
3. Remaining Decoding (Steps 10–49) & Final VQ	$\sim$ 8.92 s	Bypassed **(Saved)**
4. Full Reward Eval (BER, DER, FER)	$\sim$ 0.11 s	Bypassed **(Saved)**

Table 5: **Catastrophic Forgetting Evaluation.**

Method	**GenEval**	**MMMU**	**POPE**	**VQAv2 (COCO)**	**COCO-FID**
Base Model	0.67	26.7	80.0	0.72	8.97
UniCTokens-SFT	0.66	26.7	79.8	0.70	8.98
Sync-R1	0.65	26.5	79.8	0.72	9.01